



Toegepaste Micro-economie: Public economics

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Week 6:
Optimal Taxation

6. Optimal Taxation

Last week:

- Tax incidence
 - Who is effectively paying for the tax revenue?
 - **Distribution**
- **Efficiency** effects:
 - Taxes distort relative prices and, hence, choices
 - Result in lower efficiency: **Excess burden**

Today:

- ‘Optimal’ taxation

6. Optimal Taxation

Three main questions today:

1. If a certain amount of tax revenue has to be raised, what is the least inefficient way?
→ Minimising total Excess Burden
2. Does a proposal (intervention/redistribution) still improve SW when taking EB into account?
→ Public provision: subsidies; manpower to enforce regulation; infrastructure; etc.
3. Evaluation of tax systems
→ Efficiency *and* distributional issues

Today's roadmap

1. How to minimise Excess Burden when raising R in tax revenue?
 - Ramsey rule (or: 'Inverse elasticity rule')
 - Assumptions and derivation
 - Example
2. Taking Excess Burden into account when deciding about intervention/redistribution
 - Intuition and old exam exercise
3. Criteria for evaluating tax systems

Minimising Excess Burden

Suppose :

- We need to raise R in tax revenue
- We want to distort as little as possible
(so, we ignore distribution / assume all individuals are identical)
- It is impossible to use a lump-sum tax

So, how to set rate on different commodities / input factors such that we raise R at minimal total Excess Burden?

Minimising Excess Burden

First guess: All commodities equal tax rate?

Actually, if we could tax all commodities *including Leisure*, there would be no Excess Burden:

- Three ‘goods’: X , Y , and Leisure L
- Time endowment T , wage rate w , tax rate t

→ Budget constr. without tax: $wT = P_X X + P_Y Y + wL$

→ BC with tax: $wT = (1+t)P_X X + (1+t)P_Y Y + (1+t)wL$

This equals
$$w \frac{T}{(1+t)} = P_X X + P_Y Y + wL$$

Minimising Excess Burden

$$w \frac{T}{(1+t)} = P_X X + P_Y Y + wL$$

This is as if the tax reduces time endowment T

- Whatever the individual does, it is not possible to reduce amount of tax paid
 - No change in relative prices, so no distortions (no substitution effects)

Hence: taxing all goods including leisure at the same rate is a lump-sum tax: Excess Burden is zero

Minimising Excess Burden

However, taxing leisure is not possible

Only taxing goods X and Y gives budget constraint

$$wT = (1+t)P_X X + (1+t)P_Y Y + wL \quad (1)$$

This changes relative prices (of leisure versus X and Y)

→ leading to distortions and, hence, Excess Burden

Note:

Taxing all goods except leisure at the same rate gives the same budget constraint as a proportional income tax:

Rewriting (1) gives
$$\frac{w}{(1+t)}(T - L) = P_X X + P_Y Y$$

Minimising Excess Burden

So:

- We can focus on taxing commodities
- And we know that there will be an Excess Burden

Now the question is:

Which commodities should we tax, and at which rate?

Remember that our goals are:

- Raise R in tax revenue
- Minimise the total Excess Burden
→ No distributional concerns

Minimising Excess Burden

Two simplifying assumptions:

1. Supply curves are perfectly elastic (horizontal)
(→ in other words: constant marginal cost)
2. Cross-price elasticities are zero
→ Goods are neither substitutes nor complements

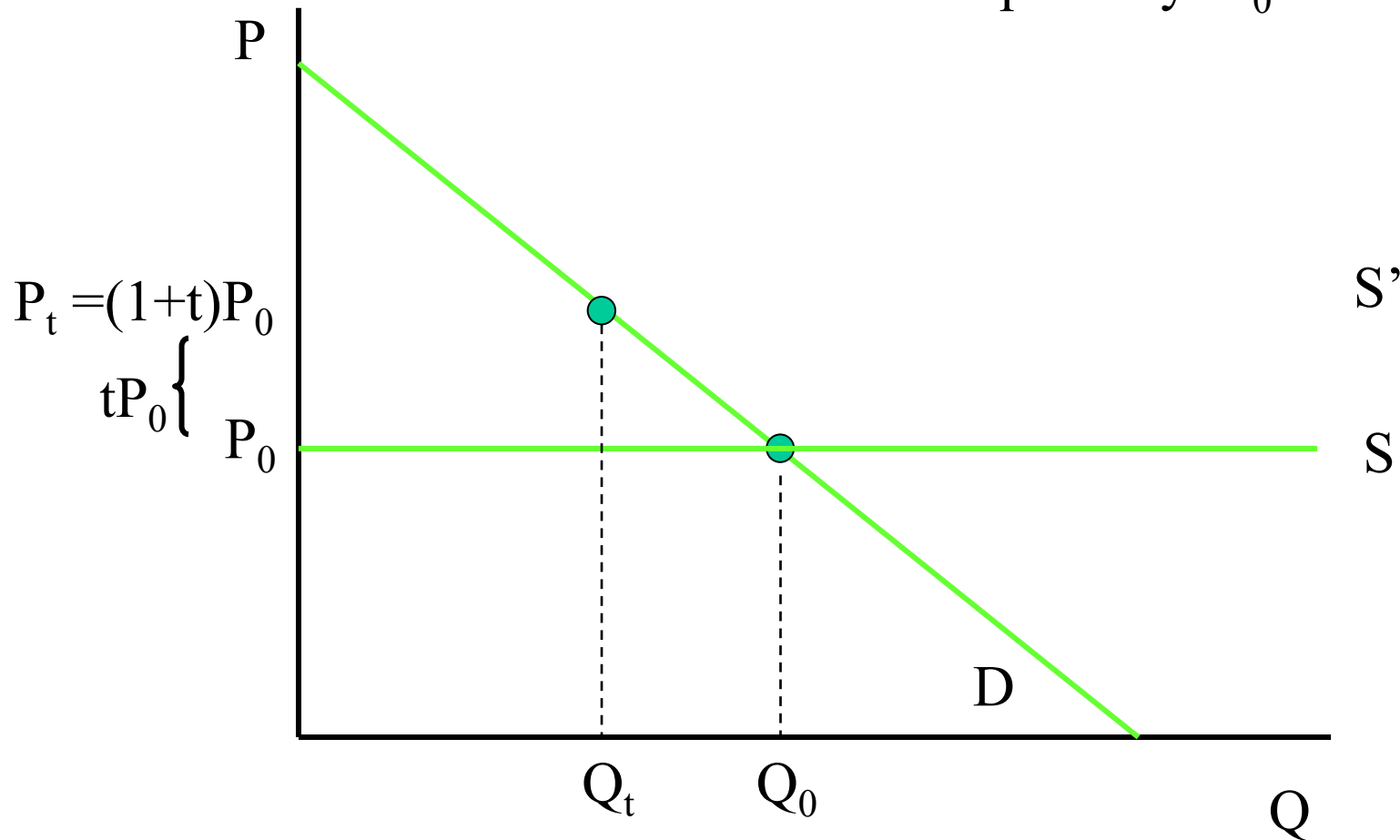
This implies

1. All effects on a market run through demand
2. The effects of a tax on a certain good are contained to the market for that good
→ substitution effect towards leisure

Minimising Excess Burden

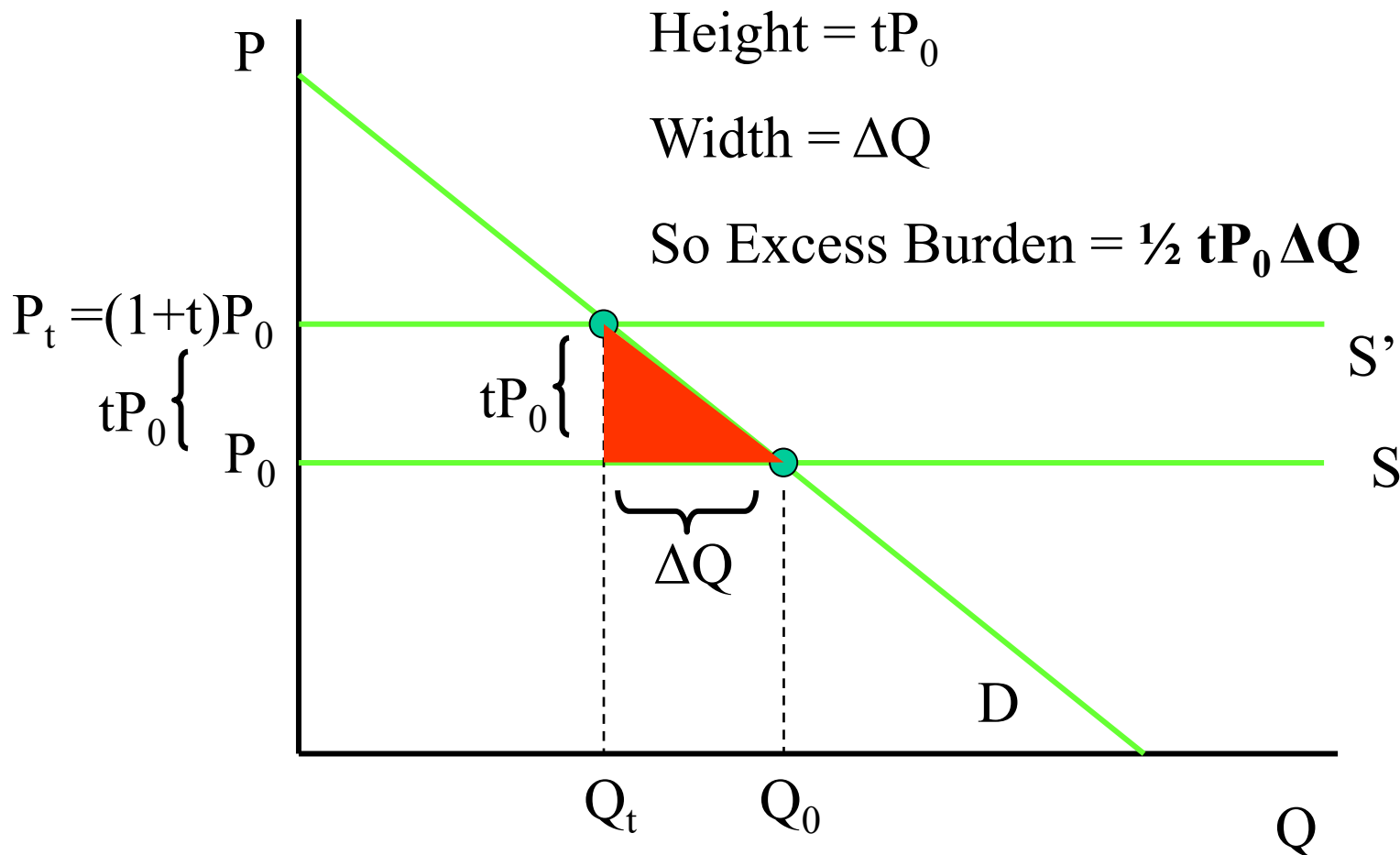
Ad valorem tax t (tax rate = t percent of producer price)

Raises consumer price by tP_0



Minimising Excess Burden

Size of the excess burden (perf. elastic supply, linear demand,
ad val. tax: Triangle:



Minimising Excess Burden

So: Excess Burden is $\frac{1}{2} tP_0 \Delta Q$

Note:

- tP_0 is the change in consumer price: ΔP
- ΔQ is the change in equilibrium quantity

Absolute value of demand elasticity in
before-tax situation:

$$\varepsilon_d = \left| \frac{\Delta Q}{\Delta P} \frac{P_0}{Q_0} \right|$$

Using $tP_0 = \Delta P$, we can rewrite this to

$$\Delta Q = \varepsilon_d t Q_0$$

Substituting this into EB gives

$$EB_t = \frac{1}{2} \varepsilon_d P_0 Q_0 t^2$$

Minimising Excess Burden

$$EB_t = \frac{1}{2} \varepsilon_d P_0 Q_0 t^2$$

Two important results:

- EB increases in demand elasticity
 - The larger the elasticity, the larger the reduction in Q after the increase in consumer price
 - Larger distortion!
- EB increases *quadratically* in tax rate t
 - The higher the tax rate, the greater the increase in the EB if the tax rate increases
 - The *Marginal Excess Burden* is increasing in t

Minimising Excess Burden

Similarly, we can find Excess Burden with unit tax u
(under perf. elastic supply)

Completely identical, except that now the (absolute)
level of tax is not tP_0 but u

This gives
$$EB_u = \frac{1}{2} \varepsilon_d \frac{Q_0}{P_0} u^2$$

Hence, also similar effects of ε and u

Minimising Excess Burden

Remember:

Goal is to raise R in tax revenue, at minimum efficiency cost

→ minimise total excess burden from tax system

From EB formula:

- EB in single market increases quadratically in tax
→ Better to spread taxes over markets than to focus on a few markets
- EB increases in elasticity
→ Lower taxes on goods with high elasticity than on goods with low elasticity

Ramsey Rule

Optimal tax rates on different commodities:

Marginal EB of the *marginal* euro tax revenue raised should be equal across commodities

→ Otherwise, a reduction in revenue raised in market with high MEB offset by an increase in revenue raised in market with lower MEB yields lower total Excess Burden

To raise R_X on the market for good X, ad val tax rate should be

$$t_X = \frac{R_X}{P_X Q_X}$$

Marginal increase in R_X yields: $\frac{\delta t_X}{\delta R_X} = \frac{1}{P_X Q_X}$

Ramsey Rule

So, marginal tax revenue raises tax by $\frac{\delta t_X}{\delta R_X} = \frac{1}{P_X Q_X}$

The change in t_X subsequently affects the Excess Burden in the market for good X

$$\text{As } EB_X = \frac{1}{2} \varepsilon_X P_X Q_X t_X^2 ,$$

the effect of an increase in t_X on EB_X is

$$\frac{\delta EB_X}{\delta t_X} = \varepsilon_X P_X Q_X t_X$$

Ramsey Rule

Combining these two results

1. Effect of marginal increase in R_X on t_X : $\frac{\delta t_X}{\delta R_X} = \frac{1}{P_X Q_X}$

2. Effect of marginal increase in t_X on EB_X

$$\frac{\delta EB_X}{\delta t_X} = \varepsilon_X P_X Q_X t_X$$

We obtain the marginal effect of R_X on EB_X

→ The **Marginal Excess Burden**

$$MEB_X = \frac{\delta t_X}{\delta R_X} \frac{\delta EB_X}{\delta t_X} = \frac{\varepsilon_X P_X Q_X t_X}{P_X Q_X} = \varepsilon_X t_X$$

Ramsey Rule

$$MEB_X = \varepsilon_X t_X$$

So, MEB increases in elasticity and in the tax rate

Recall: total Excess Burden is minimal when the
Marginal Excess Burdens of all goods are equal

So, for any two goods X and Y, it must be that:

$$MEB_X = MEB_Y$$

Which gives: $\varepsilon_X t_X = \varepsilon_Y t_Y$

Or:

$$\frac{t_X}{t_Y} = \frac{\varepsilon_Y}{\varepsilon_X}$$

Ramsey Rule

So, total excess burden is minimal when tax rates are such that for any two goods, we have

$$\frac{t_X}{t_Y} = \frac{\varepsilon_Y}{\varepsilon_X}$$

→ Ramsey rule, or ‘Inverse elasticity rule’

Ratio of tax rates is inversely proportional to ratio in elasticities

→ The higher elasticity, the lower the tax rate

Ramsey Rule

Current Dutch practice:

Most goods have VAT-tax of 21%, except for some primary goods (like food), which have 9% VAT

Is this in line with the Ramsey rule?

Ramsey Rule

Side note (see also footnote 4, ch. 16):

Ramsey rule easily derived by using Lagrange:
(optimisation under constraint)

- Objective: Minimise total Excess Burden
- Constraint: Raise R in tax revenue

With two goods X and Y, this gives Lagrange equation

$$\Gamma = \frac{1}{2} \varepsilon_X P_X Q_X t_X^2 + \frac{1}{2} \varepsilon_Y P_Y Q_Y t_Y^2 + \lambda [t_X P_X Q_X + t_Y P_Y Q_Y - R]$$

Which yields the R. Rule in the process of optimisation

Government Intervention and Excess Burden

What are the implications for the fact that raising taxes lead to reductions in efficiency?

Answer is simple:

Excess Burden is a cost, and should therefore be taken into account in a cost-benefit analysis

Example: Betuwelijn

- Construction cost: 4.7 billion euro
 - Economic cost: 4.7 bn + EB of raising 4.7 bn euro
- (Benefits: unknown)

Government Intervention and Excess Burden

Similarly: Level of subsidy on education

- Optimal level: marginal benefits of last euro of subsidy equal to marginal cost
- MB: higher efficiency because of pos. ext. (?)
- MC: $1 + \text{Marginal Excess Burden}$ of raising one more euro of tax revenue
- Referred to as

Marginal Cost of Public Funds (MCPF)

Government Intervention and Excess Burden

Estimates of the MCPF:

- For OECD countries, on average 1.2 – 1.3
- Prof. Bas Jacobs (2008): 1.5 for the Netherlands

Estimates depends on (labour) elasticities and tax levels

Note:

Also relevant for redistribution: MCPF of 1.5 implies that the last redistributed euro reduces efficiency by 50 cents.

Practice:

No (systematic) attention for (Marginal) Excess Burden in public decision-making

Government Intervention and Excess Burden

Qualifying insight by Jacobs (2018):

Governments can always use a lump-sum tax to raise (marginal) tax revenue.

→ Good for efficiency, ‘bad’ for redistribution

In practice, governments use distortive taxes

Apparently, the redistributive gains of these distortive taxes offset the marginal excess burden (assuming a SW-maximising government)

Evaluation of tax systems

Criteria for evaluation tax systems:

(Borrowed from Gregory Mankiw: gregmankiw.blogspot.com)

1. **Efficiency**: minimise distortion of individ. choices
(Or in case of externalities: correct these choices)
2. **Egalitarianism**: more equal after-tax income distribution
(ethical point of view → SW-function)
3. **Intergenerational equity**: revenue should be such that current generation does not burden future generations too much
→ government debt; saving for aging of current generation
- (4. **Stabilisation**: dampening of the business cycle
→ Keynesian policies)

Evaluation of tax systems

Big debate about the 4th point:

Can the government stabilise the economy?

Examples:

- 2011 US budget deficit: **\$1.500.000.000.000** , hoping to stimulate the economy
- 2020 US budget deficit: **\$3.100.000.000.000**, due to Covid
→ 14.9% of GDP
- 2020 NL budget deficit: **€34.000.000.000**, due to Covid
→ 4.3% of GDP
- 2022 – 2023 UK budget ‘plans’: deficit of **72,000,000,000** pounds in period of 6 months, due to energy crisis and ...
→ 6.4% of GDP

Evaluation of tax systems

Proponents of stabilisation policies:

- Timely increases and decreases in government net expenditures help to dampen the business cycle
- Helps to maintain production capacity

Opponents:

- In practice, governments often act too late, which makes that policies may actually strengthen the business cycle
- Politicians do tend to increase net exp. in 'bad' times, but do not decrease net exp. in 'good' times
- Risk of undertaking projects that fail cost-benefit analysis

Evaluation of tax systems

Criteria 1, 2, and 3 give rise to the, by now familiar,

Trade-off between efficiency and distribution

Some policies good for efficiency, bad for distribution

→ fat tax; energy tax; strict benefit eligibility rules

Other policies bad for efficiency, good for distribution

→ Progressive income tax; low tax rate on food

Evaluation of tax systems

Criteria 1, 2, and 3 give rise to the, by now familiar,

Trade-off between efficiency and distribution

Some policies greatly affect intergenerational distribution

→ Pay-as-you-go public pensions; no government ‘saving’ for increasing social sec. & health care cost due to aging;

Results: You will pay the bill of rising public expenditures to people currently enjoying ‘low’ tax rates

Related: discussion on eligibility age for (public) pensions

→ (Absence of) climate change policies

Efficiency: How to best internalize this externality?

Distribution: How to divide cost of (preventing) climate change within and across generations?

Next week

So far:

Normative analysis of government intervention

→ When / how **should** the government intervene?

Next week:

Positive analysis of government intervention

→ How are decisions being made:

Collective decision-making

→ Direct democracy

→ Representative democracy